

veriroute

Verification-driven, token-efficient routing agent

Verified answers cost **zero tokens**. Only proven failures escalate to Fireworks.

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github.com/bogdan-lmk/veriroute

THE PROBLEM

Fewest tokens, above the accuracy gate

- > Scoring: pass an LLM-judge accuracy gate, then **ascending total Fireworks tokens** — every wasted token is rank lost
- > Reasoning models bill **hidden thinking as output tokens** — a tight cap returns empty answers, a loose one burns budget
- > Prompt-based routers pay for a classification call on **every single task**
- > Half the field never scored at all: packaging, timeouts, malformed output

THE APPROACH

Harness engineering, not prompt magic

- > **Deterministic policy table** routes every task — the LLM never decides its own escalation
- > **Verification by execution**: math via program-of-thought, codegen via generated self-tests — local answers ship only when code proves them
- > **Token-oriented model ranking**: non-thinking models first, accuracy costs the same capped tokens; Gemma ranks first when allowed
- > Local inference is **zero tokens** — llama.cpp CPU-dispatch runtime already packaged in the image

RELIABILITY BY CONSTRUCTION

Every failure mode maps to a guardrail

- > Stub-first **atomic results.json** before anything can fail; rewritten after every task — SIGKILL-safe (integration-tested)
- > `model in ALLOWED_MODELS` asserted **before any network I/O** — disqualification is structurally impossible
- > Hard token budget stop · 28s per-task cap · parallel drain at T-45s · exit 0 in every failure mode
- > **49 tests**; one change per submission, predictions logged vs leaderboard

MEASURED LIVE ON FIREWORKS (MINIMAX-M3, 8 PRACTICE TASKS)

Escalation output policy, chosen by data

Config	Tokens	Answers intact
cap 300, no instruction	3,065	5/8 — truncated mid-answer
terse + thinking cap 600	4,411	8/8, but paid retry burn
terse + thinking cap 1000	3,573	8/8

Terse suffix cuts simple-task completions 40–60%; thinking-aware caps end the empty-answer retry loop.

ROADMAP

Every verified category → zero tokens

- > Local Qwen-class model behind code verification takes sentiment, NER, summarization, math & codegen off the paid path
- > Escalation remains only for proven failures and factual recall — bounded by the token budget
- > Target: **beat 5,121 tokens** (current sole gate-passer) with a wide accuracy margin

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